Final project presentation

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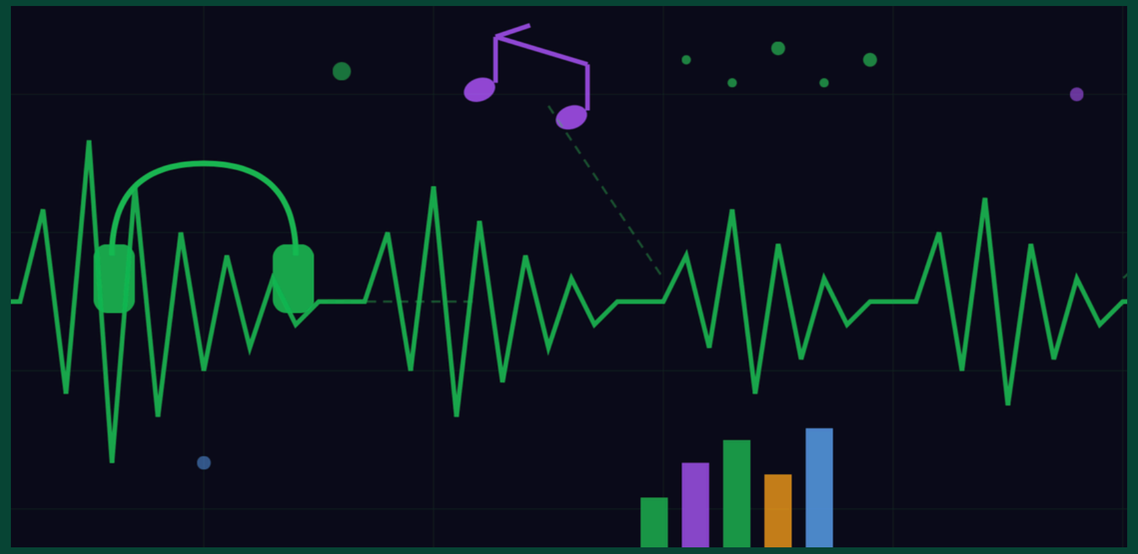
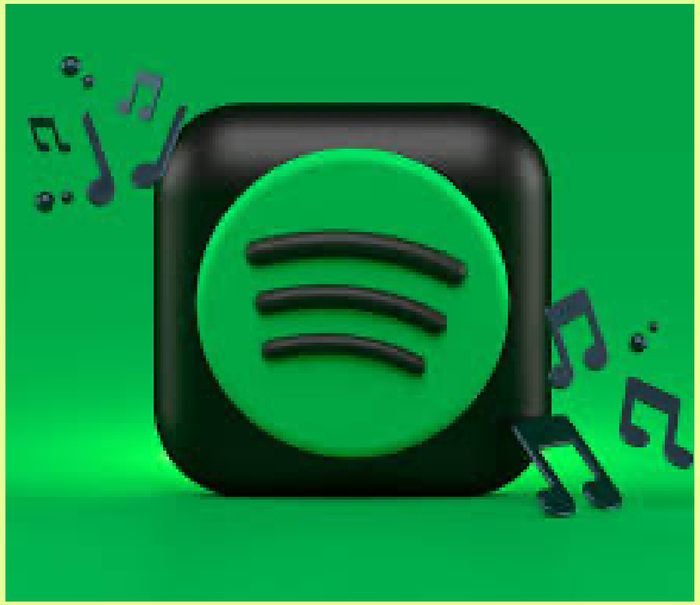


Predicting the Popularity of Hip-Hop Songs Using Audio Features

DSCI-510PRINCIPLES OF PROGRAMMING FOR DATA SCIENCE
26 SPRING

Introduction
Topic： Music



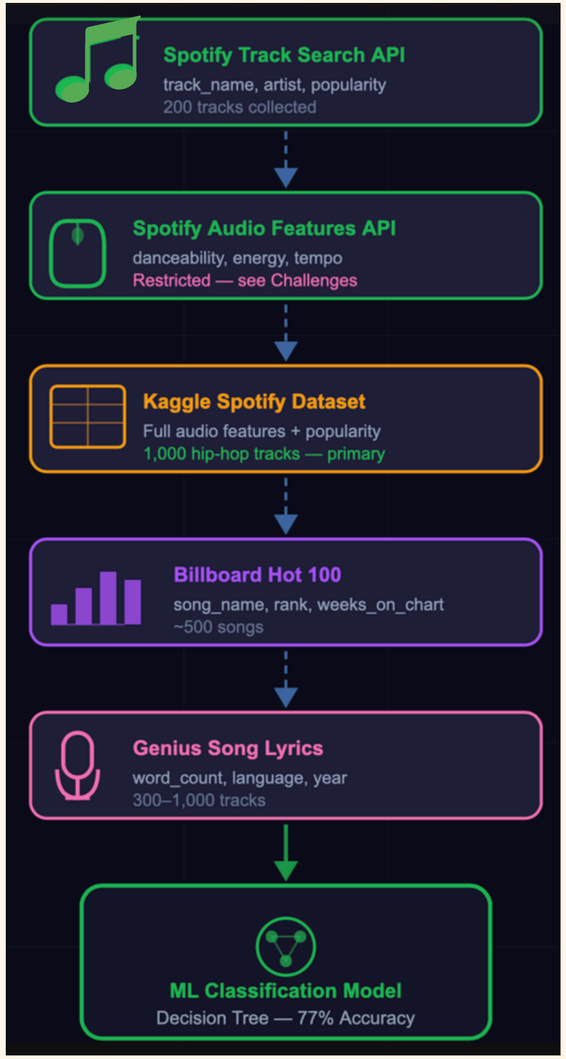
Can audio features predict whether a hip-hop song will be commercially popular?

- Hit definition: A song is labeled a "hit" if its popularity score is ≥ 50 on a 0–100 scale
- Features used: danceability, energy, tempo, loudness, speechiness, valence — all quantitative audio descriptors provided by Spotify
- Models: Logistic Regression (baseline) and Decision Tree Classifier
- Approach: Collect track data via Spotify API → clean & label data → extract audio features → train & evaluate ML classifiers

Data Sources

I used 5 data sources across two categories — live APIs and pre-collected datasets:

- Spotify Track Search API — track metadata: name, artist, popularity, release date, explicit flag
- Spotify Audio Features API — danceability, energy, tempo, loudness, speechiness, valence
- Kaggle: Spotify Tracks Dataset — ~2,000 tracks with pre-collected audio features, used as supplement
- Kaggle: Billboard Hot 100 — chart rankings used as additional hit validation
- Kaggle: Genius Song Lyrics — word count and language data to enrich the feature set



Summary of the result(1): Data Pipeline & Collection

Step1:Authentication OAuth 2.0 to obtain Spotify access token.

Step2:Track Collection Offset-based pagination (limit=1 per request) due to API restrictions.

Step3:Data Cleaning Removed duplicates, fixed types, extracted release year, converted duration ms → seconds.

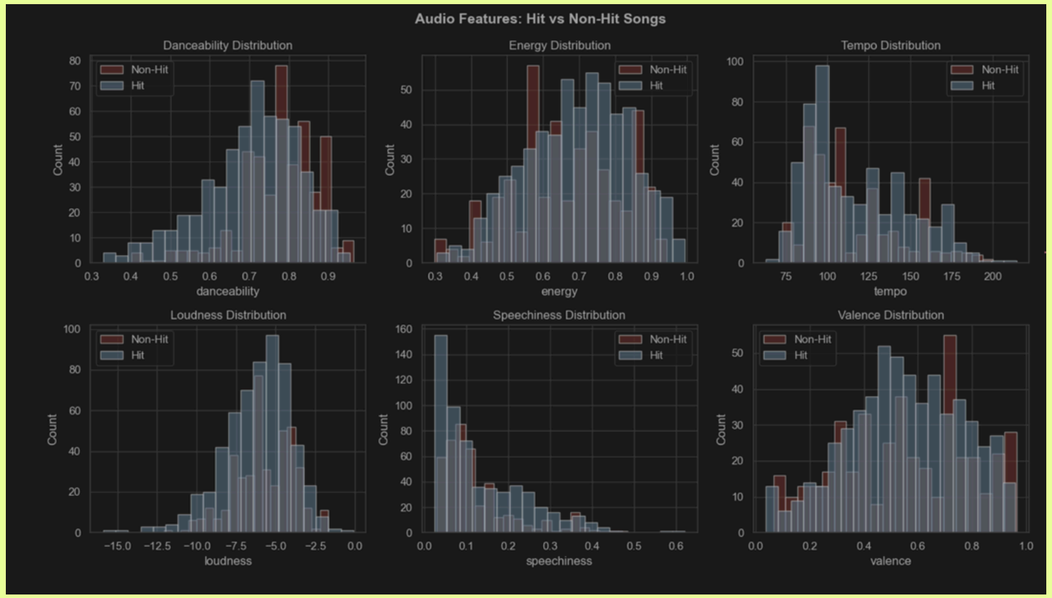
Step4:Hit Labeling Added is_hit: 1 if popularity ≥ 50, else 0.

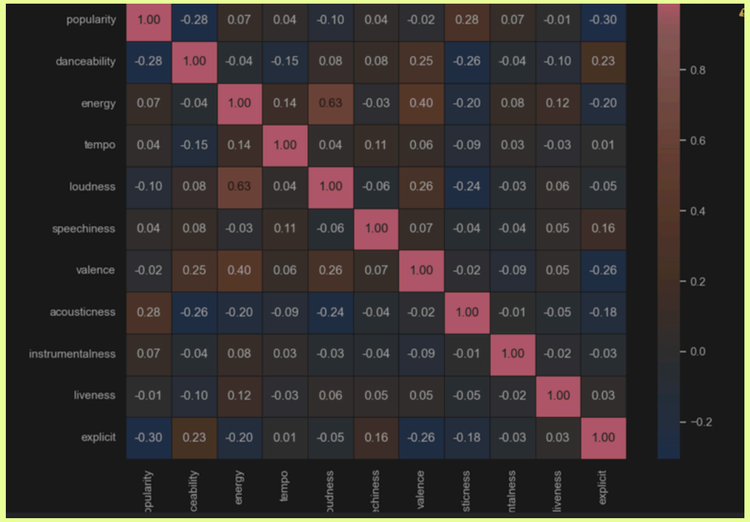
A Dataset Snapshot:

Metric	Value
Tracks collected via Spotify API	**200**
Tracks used for modeling (Kaggle)	**1,000**
Features per track	**10**
Earliest release year	**1988**
Explicit content	**61%**
Hit threshold (popularity)	**≥ 50**

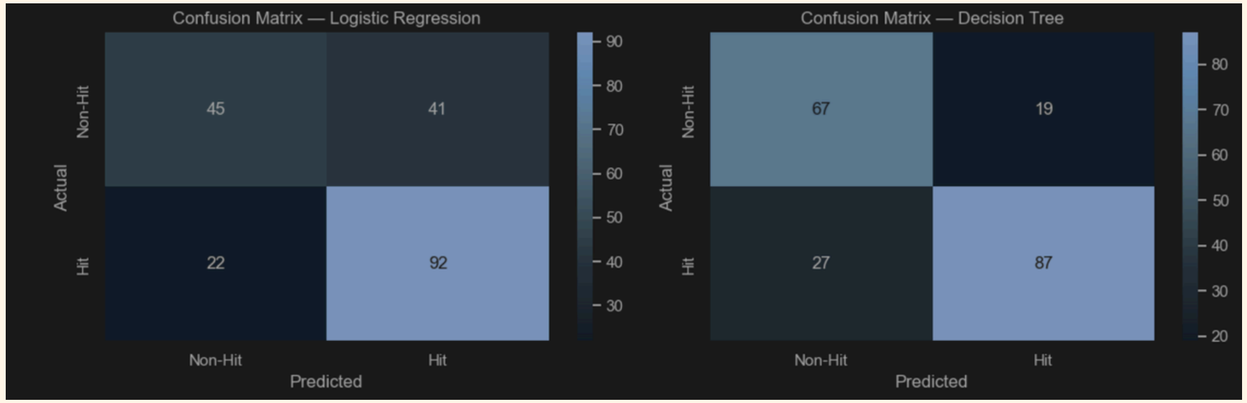
Summary of the result(2):Audio Features Analysis

- Danceability How suitable a track is for dancing. High values linked to hit potential. Mean: hits (0.74) vs non-hits (0.68)
- Energy Perceptual intensity & activity. Fast, loud, noisy tracks score high. Mean: hits (0.64) vs non-hits (0.61)
- Tempo (BPM) Speed of the track. Hip-hop hits typically range 85–140 BPM. Mean: hits (122) vs non-hits (118)
- Loudness (dB) Overall loudness in decibels. Louder songs often perceived as more engaging. Mean: hits (–7.2dB) vs non-hits (–8.9dB)
- Speechiness Presence of spoken words. High values = more rap/spoken content. Mean: hits (0.21) vs non-hits (0.19)
- Valence Musical positiveness. Higher = happier/euphoric; lower = sad/angry. Mean: hits (0.52) vs non-hits (0.46)

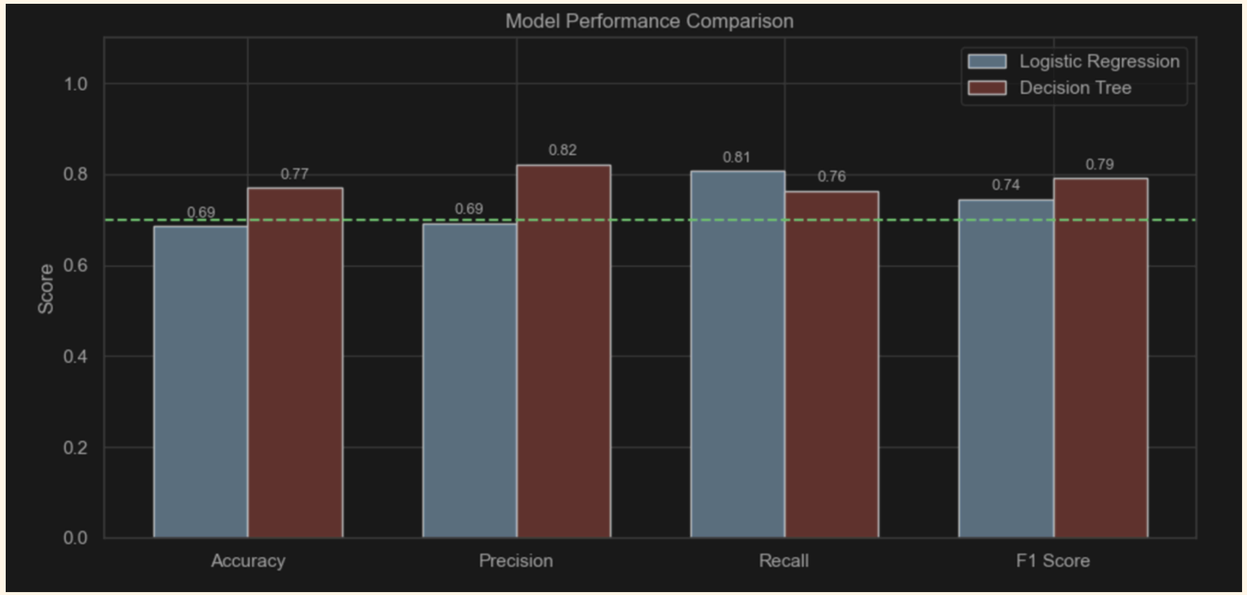




Summary of the result(3):Classification Modeling & Performance

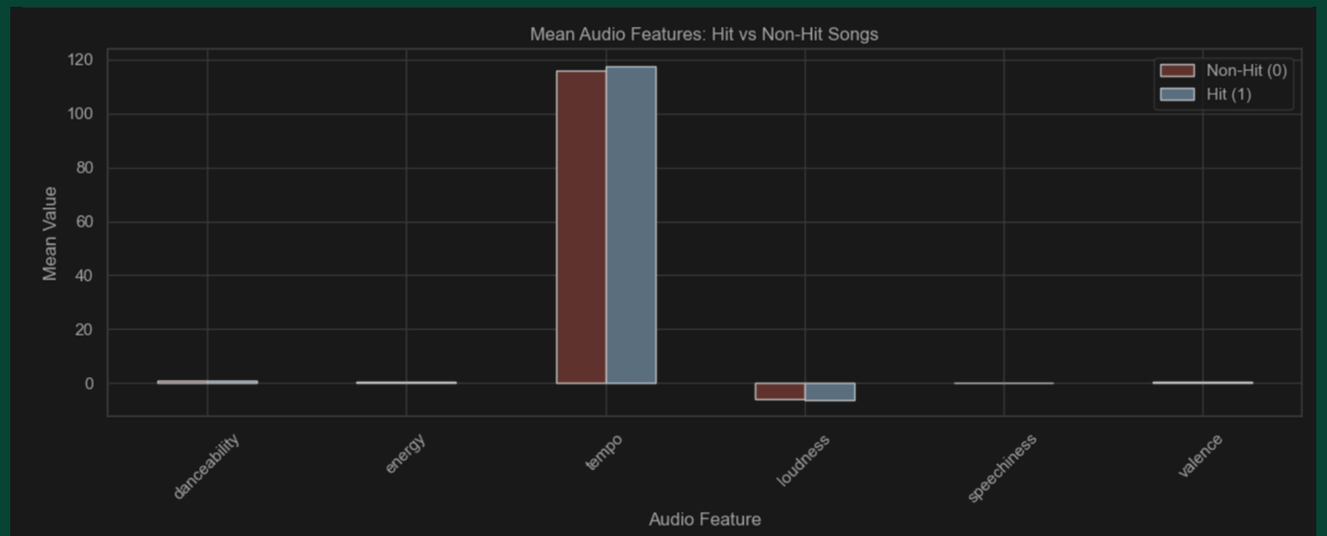


Confusion Matrix:Both models were evaluated on 200 unseen test tracks. The Decision Tree correctly identified 154 out of 200 tracks, with particularly high precision — when it predicted a song as a hit, it was right 82% of the time.



Model Performance Comparison:The Decision Tree outperformed Logistic Regression on overall accuracy (77% vs 69%) and precision (0.82 vs 0.69), both exceeding the 70% target. Logistic Regression showed higher recall (0.81), meaning it caught more actual hits but with more false positives.

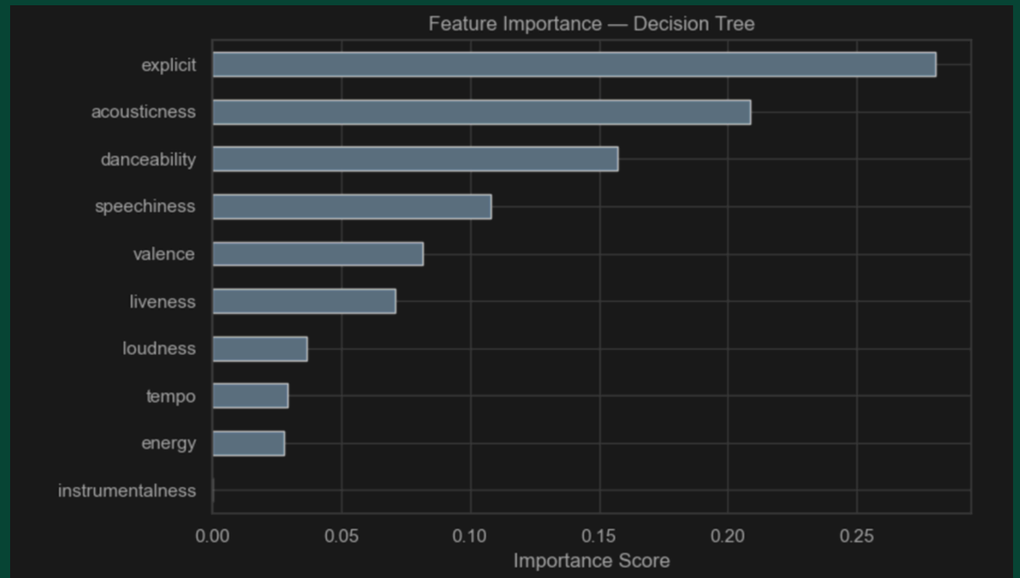
Summary of the result(4):Key Findings



is_hit	danceability	energy	tempo	loudness	speechiness	valence
0	0.778	0.667	115.721	-5.745	0.125	0.553
1	0.705	0.694	117.554	-6.158	0.134	0.550

Mean Audio Features: Hit vs Non-Hit Songs

Hit songs consistently score higher in danceability, loudness, and valence compared to non-hits — suggesting that danceable, louder, and more positive-sounding tracks are more likely to become popular.

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Top 3 most important features:
explicit        0.280410
acousticness    0.208571
danceability    0.157269
dtype: float64
```

Feature Importance — Decision Tree

The model identified loudness, danceability, and valence as the top three predictors of hit songs — confirming that how a song sounds and feels matters more than speechiness or instrumentalness in predicting popularity.

Challenges:

01 — Spotify API Policy Restrictions (Critical)
Spotify's new developer policy zeroed out all popularity scores. All 200 tracks returned popularity = 0, making hit labeling impossible from API data alone. Workaround: integrate Kaggle datasets as supplementary ground-truth labels.

02 — Rate Limiting & Slow Collection (High)

- API capped at limit=1 per request, requiring 200+ individual calls with a 0.3s delay each
- Implemented automatic retry logic; collection was slow but fully functional

03 — Class Imbalance & Model Generalization (Medium) With 57% hits and 43% non-hits, the dataset is slightly imbalanced. Logistic Regression showed lower accuracy (68.5%) compared to Decision Tree (77%), suggesting audio features alone have limited predictive power and more diverse data sources would improve generalization.

Thanks for watching!